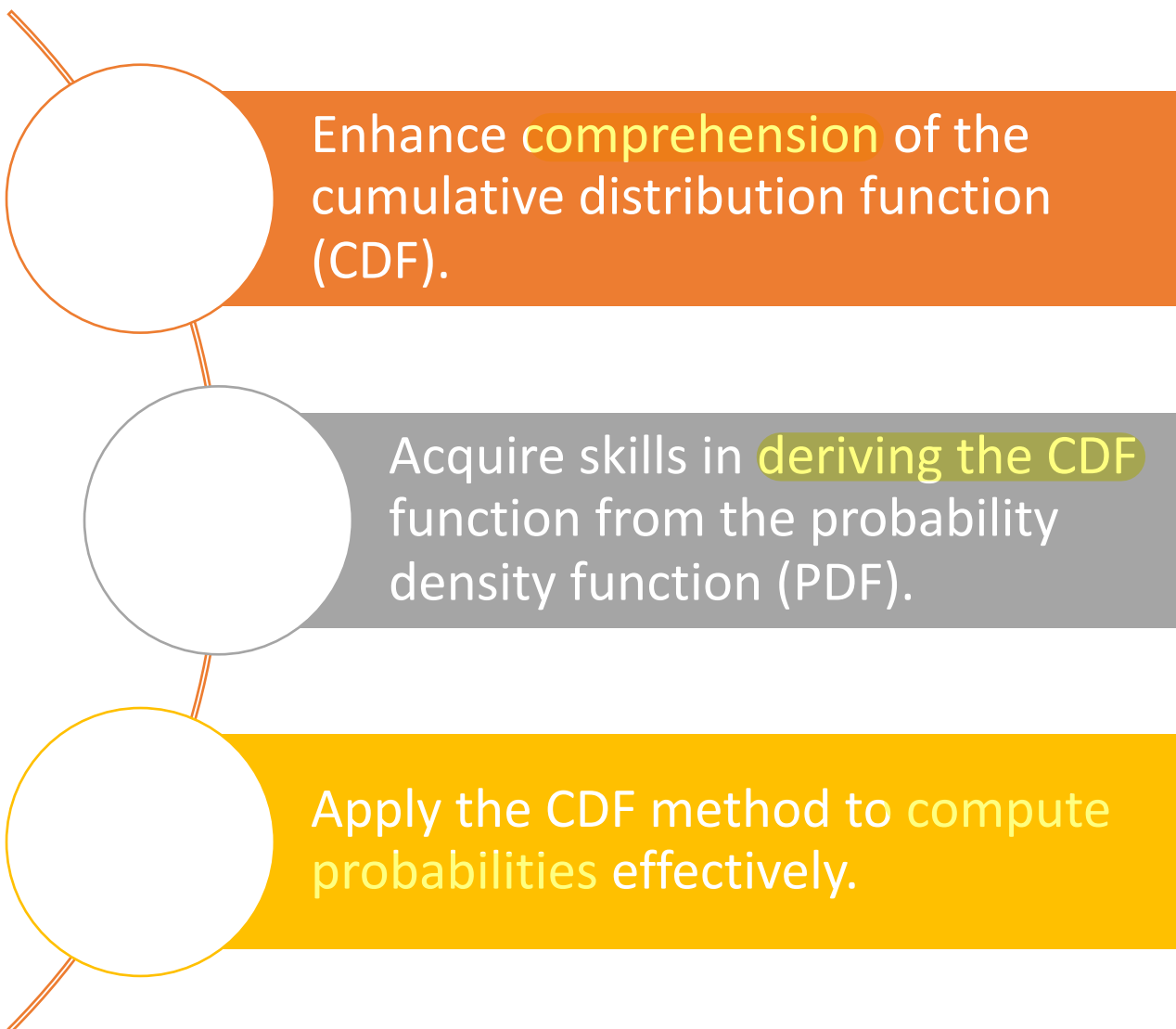




Stat 3113: Lesson 15

Cumulative Distribution Function
for Continuous Random Variables

Learning Objectives



Enhance **comprehension** of the cumulative distribution function (CDF).

Acquire skills in **deriving the CDF** function from the probability density function (PDF).

Apply the CDF method to **compute probabilities** effectively.

Cumulative Distribution Functions

- The cdf F of a continuous random variable has the same definition as that for a discrete random variable. That is,

$$F(x) = P(X \leq x)$$

- In practice this means that F is essentially a particular antiderivative of the pdf since

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$$

- Thus at the points where f is continuous $F'(x)=f(x)$.

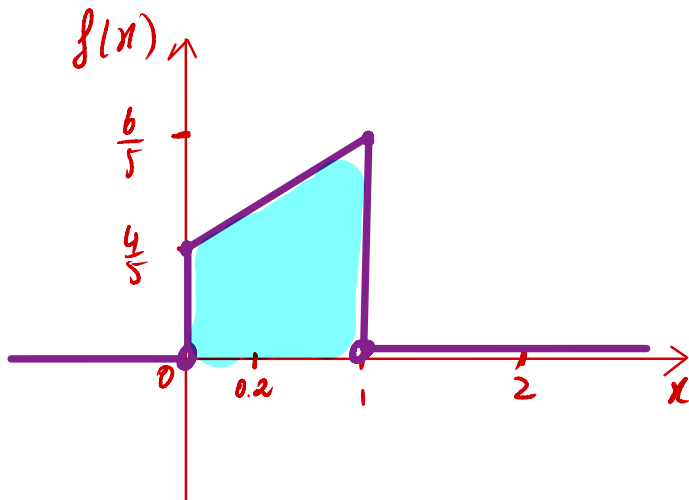
Probability Formulas for Continuous Random Variables

1. $P(X < a) = P(X \leq a) = F(a)$ this is the area under the curve of the pdf, $f(x)$, x less than a .
2. $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b) = F(b) - F(a)$
this is the area under the curve of the pdf, $f(x)$, x between a and b .
3. $P(X > a) = P(X \geq a) = 1 - F(a)$ this is the area under the curve of the pdf, $f(x)$, x greater than a .

Example 1: The proportion of people who respond to a certain mail-order solicitation is a continuous random variable X that has the following probability density function:

$$f(x) = \begin{cases} \frac{2}{5}(x+2) & , 0 \leq x \leq 1 \\ 0 & , \text{ow} \end{cases}$$

a. Graph the pdf, and then find the cdf.



$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du$$

$$\cdot \text{ For } x < 0, F(x) = \int_{-\infty}^x 0 du = 0$$

$$\cdot \text{ For } 0 \leq x < 1, F(x) = \int_{-\infty}^x f(u) du = \underbrace{\int_{-\infty}^0 f(u) du}_0 + \int_0^x f(u) du = \int_0^x \frac{2}{5}(u+2) du$$

$$= \frac{2}{5} \left[\frac{1}{2} u^2 + 2u \right]_0^x = \frac{2}{5} x \left(\frac{1}{2} x + 2 \right) = \frac{1}{5} x(x+4)$$

$$\cdot \text{ For } x \geq 1, F(x) = \int_{-\infty}^x f(u) du = \underbrace{\int_{-\infty}^0 f(u) du}_0 + \underbrace{\int_0^1 f(u) du}_0 + \int_1^x f(u) du = \int_0^1 f(u) du = 1$$

Example 1: More Space for Work

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{x}{5}(x+4) & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

b. Use cdf to find: $P(0.5 < X \leq 0.75)$

$$\begin{aligned} P(0.5 < X \leq 0.75) &= F(0.75) - F(0.5) \\ &= 0.7125 - 0.45 \\ &= 0.2625 \end{aligned}$$

c. Use cdf to find: $P(X > 0.65)$

$$\begin{aligned} P(X > 0.65) &= 1 - P(X \leq 0.65) \\ &= 1 - F(0.65) \\ &= 0.3955 \end{aligned}$$

d. Use cdf to find: $P(X \leq 0.2)$

$$P(X \leq 0.2) = F(0.2) = 0.168.$$

Example 2: Let X be a continuous random variable with pdf $f(x) = \begin{cases} \frac{3}{2} \cdot \frac{1}{x^2} & , 1 \leq x \leq 3 \\ 0 & , \text{ow} \end{cases}$

Find the cdf.

$$F(x) = P(X \leq x)$$

- For $x < 1$, $F(x) = \int_{-\infty}^x f(u) du = \int_{-\infty}^x 0 du = 0$

- For $1 \leq x \leq 3$, $F(x) = \int_{-\infty}^x f(u) du = \underbrace{\int_{-\infty}^1 f(u) du}_0 + \int_1^x f(u) du = \int_1^x \frac{3}{2} \cdot \frac{1}{u^2} du = \frac{3}{2} \left[-\frac{1}{u} \right]_1^x$

$$= \frac{3}{2} \left(1 - \frac{1}{x} \right)$$

- For $x > 3$, $F(x) = \int_{-\infty}^x f(u) du = \underbrace{\int_{-\infty}^1 f(u) du}_0 + \underbrace{\int_1^3 f(u) du}_0 + \int_3^x f(u) du = \int_1^3 \frac{3}{2} \frac{1}{u^2} du = 1$

$$F(x) = \begin{cases} 0 & \text{if } x < 1 \\ \frac{3}{2} \left(1 - \frac{1}{x} \right) & \text{if } 1 \leq x \leq 3 \\ 1 & \text{if } x > 3 \end{cases}$$

Example 2: Let X be a continuous random variable with pdf

$$f(x) = \begin{cases} \frac{3}{2} \cdot \frac{1}{x^2} & , 1 \leq x \leq 3 \\ 0 & , \text{ow} \end{cases}$$

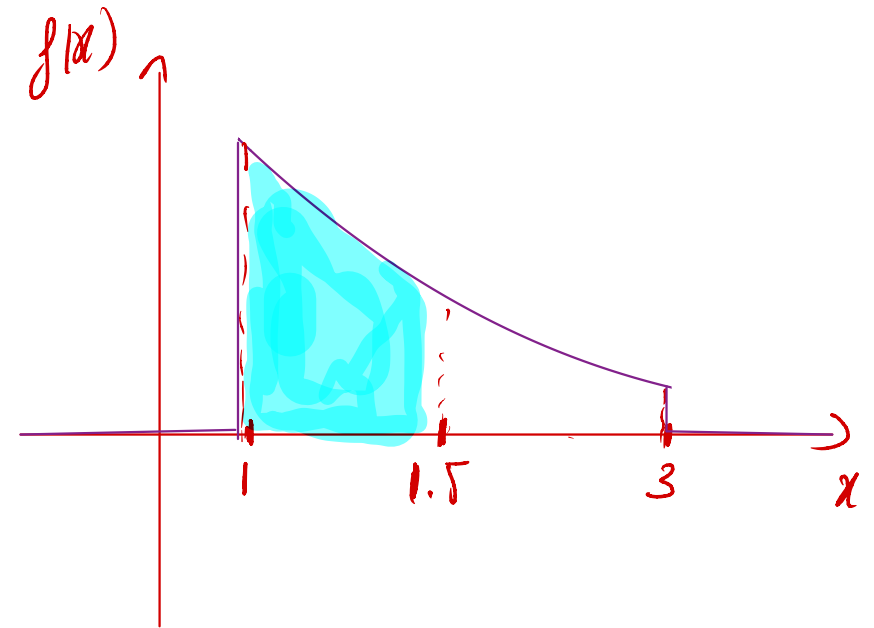
Find the median of the distribution.

$$P(X \leq x_m) = F(x_m) = 0.5$$

$$\Rightarrow \frac{3}{2} \left(1 - \frac{1}{x_m}\right) = 0.5$$

$$\Rightarrow 1 - \frac{1}{x_m} = \frac{2}{3} \cdot \frac{1}{2}$$

$$\Rightarrow \boxed{x_m = \frac{3}{2} = 1.5}$$



Reading

Chapter 3 – Section 3.3



Next Topic



The **Exponential** Distribution



Chapter 4 - **Section 4.5**